inclusionAI / Ling-3.0-flash-fp4

Text Generation | Safetensors | bailing_hybrid | conversational | custom_code | 8-bit precision | fp8 | License: mit

🤗 Hugging Face | 🤖 ModelScope | 🐙 OpenRouter

Introduction

We're introducing Ling-3.0-flash, our next-generation native hybrid reasoning model. Operating with **124B** total and **5.1B** active parameters (~12.4% and ~8.1% of our previous 1T-class flagship Ring-2.6-1T), Ling-3.0-flash matches or outperforms its predecessor across key benchmarks.

Key highlights of the model are summarized below:

- **Native Hybrid-Linear Architecture:** Ling-3.0 adopts a native hybrid linear attention architecture from the very start of pretraining

Downloads last month
3,425

Safetensors

Model size | 66B params

Tensor type | F32 · BF16 · F8_E4M3 · I8

Chat template | Files info

Inference Providers NEW

Text Generation

This model isn't deployed by any Inference Provider.

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Collection including inclusionAI/...

Ling 3.0 Collection

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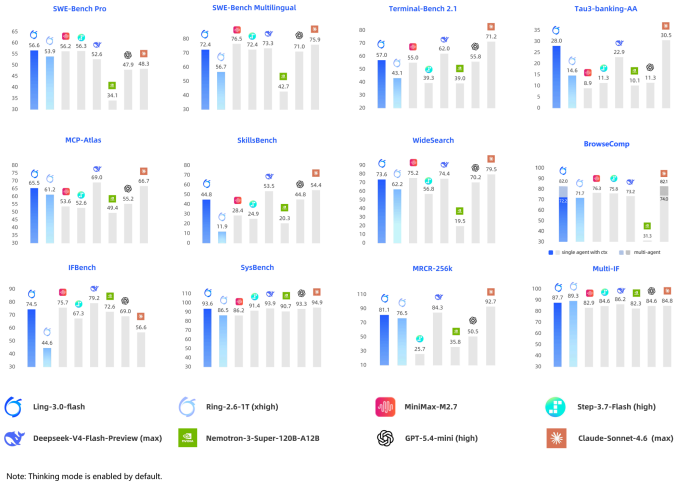
(5:1 alternating stacking of Kimi Delta Attention (KDA) and MLA), upgraded with KDA fine-grained diagonal gating and 1/64 sparse MoE. With 124B total parameters and 5.1B activated parameters, it achieves a synergistic leap in long-context efficiency and computational cost.

- **Remarkable Efficiency & Performance:** Engineered for speed, compute efficiency, and production deployment, Ling-3.0-flash delivers class-defying performance against both larger SOTA competitors and previous-generation flagships. Activating only 5.1B parameters per token, it provides impressive reasoning, instruction following, and long-context capabilities to empower complex agentic workflows in production environments.
- **Comprehensive Agentic Evolution:** Tailored for real-world productivity workflows, the model incorporates 10,000+ interactive training environments to achieve end-to-end closed-loop execution across Coding, General, and Deep Research Agent tasks. It natively integrates the SGLang HiCache + Mooncake hierarchical caching architecture (featuring physical dual-pools and a cluster-shared L3 cache), eliminating redundant recomputation during long-horizon interactions and reducing Time to First Token (TTFT) by 60% to over 80% in long-input scenarios.

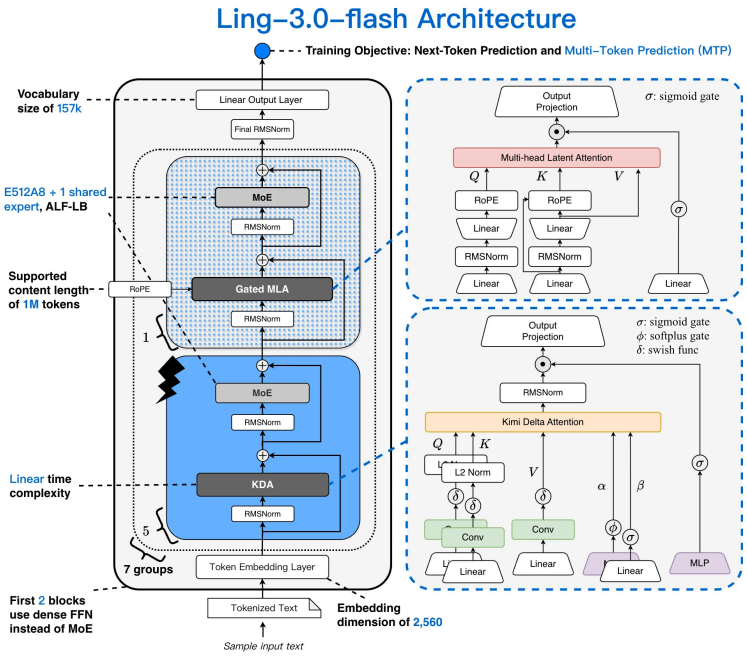


Model Overview

The model summary information and architecture diagram are as follows:

Architecture	Hybrid-linear MoE
Parameter Scale	Total 124B, Activated 5.1B
Transformer Layers	35 KDA + 7 Gated MLA (5:1)
Number of Dense Layers	2
Number of Routed Experts	512
Number of Shared Experts	1
Number of Activated Experts	8
Attention Heads	32
Hidden Size	2560
Expert Intermediate Size	768
Dense Intermediate Size	6144
Vocabulary Size	157184

Architecture	Hybrid-linear MoE
Context Training Schedule	8K -> 32K -> 256K



Evaluation

We have conducted a comprehensive evaluation of Ling-3.0-flash across multiple authoritative benchmarks. **Ling-3.0-flash** performs strongly on representative code/agent benchmarks such as **SWE-Bench Pro, SWE-Bench Multilingual, Tau3-banking-AA**, **MCP-Atlas** and **SkillsBench, etc**. In practice, Ling-3.0-flash delivers a strong user experience across frameworks including **Claude Code**,**Kilo Code**,**Qwen Code**,**Hermes Agent**,and **OpenClaw**, etc. Beyond agentic tasks, Ling-3.0-flash also delivers strong performance across **general knowledge**,**mathematical reasoning**,**instruction following**,and **long-context understanding**.

	Ling-3.0-flash	Ring-2.6-1T (xhigh)	MiniMax-M2.7	Step-3.7-Flash (high)	Deepseek-V4-Flash-Preview (max)	Nemotron-3-Super-120B-A12B	GPT-5.4-mini (high)	Claude-Sonnet-4.6 (max)
Size	124B-A5.1B	1T-A63B	230B-A10B	198B-A11B	284B-A13B	120B-A12B	-	-
Coding Agent								
SWE-Bench Pro	56.6	53.9	56.2	56.3	52.6	34.1	47.9	48.3
SWE-Bench Multilingual	72.4	56.7	76.5	72.4	73.3	42.7	71.0	75.9
Terminal-Bench 2.1	57.0	43.1	55.0	39.3	62.0	39.0	55.8	71.2
ArtifactsBench	77.0	65.1	55.8	59.2	64.0	51.6	66.8	68.7
MiniAppBench	25.3	19.2	20.7	28.0	14.8	5.8	58.8	46.3
AntSWEBench	52.2	46.4	-	-	48.6	-	-	-
General Agent								
Tau3-banking-AA	28.0	14.6	8.9	11.3	22.9	10.1	11.3	30.5
MCP-Atlas	65.5	61.2	53.6	52.6	69.0	49.4	55.2	66.7
SkillsBench	44.8	11.9	28.4	24.9	53.5	20.3	44.8	54.4
BFCL-v4	73.0	64.8	63.6	65.4	59.5	60.6	68.3	73.1
GDPVal v2-AA	1107	920	1159	1017	1189	699	-	1377
Search Agent								
WideSearch	73.6	62.2	75.2	56.8	74.4	19.5	70.2	79.5
BrowseComp	72.2 (w/ ctx) 82.0 (MA)	71.7	76.3	75.8	73.2	31.3	-	74.0 (w/ ctx) 82.1 (MA)
Draco	70.4	-	66.8	-	71.3	-	61.3	75.8
Instruction Following								
IFBench	74.5	44.6	75.7	67.3	79.2	72.6	69.0	56.6
SysBench	93.6	86.5	86.2	91.4	93.9	90.7	93.3	94.9
LIFEBench	77.3	72.5	66.9	71.3	74.1	60.2	69.2	71.8
Reasoning								
AIME26	93.2	95.8	94.2	95.0	96.5	91.7	92.9	94.4
HMMT-Feb26	87.0	93.5	71.9	87.9	94.8	84.9	83.9	85.6
IMO-AnswerBench	83.7	86.1	66.9	-	87.0	77.0	74.5	82.1
HLE	22.7	18.3	28.1	19.9	34.8	18.3	20.6	30.0
LiveCodeBench (2408-2505)	82.8	87.0	75.6	78.1	91.6	78.7	80.8	83.7
Long Context & Multi-Turn Dialogue								
MRCR_128K	90.8	90.1	27.7	39.2	88.5	40.8	56.1	92.5
MRCR_256k	81.1	76.5	-	25.7	84.3	35.8	50.5	92.7
AA-LCR	65.1	64.3	68.7	63.7	63.0	58.3	63.4	70.7
Multi-IF	87.7	89.3	82.9	84.6	86.2	82.3	84.6	84.8

- *Thinking mode is enabled by default. Unless otherwise specified, the default parameters for Ling-3.0-flash are as follows: `temperature=0.6, top_p=0.95, top_k=20`.*
- *SWE-Bench Series: Evaluated using OpenHands as the agent harness with tailored prompts. Decoding uses `temperature=0.6, top_p=0.95, max_new_tokens=32K`, with a 256K context window.*
- *Terminal-Bench 2.1: Evaluated under the Artificial Analysis (AA) protocol using the default Terminus 2 harness, a unified 2-hour timeout, the provided JSON parser in preserve-thinking mode, and 3 runs per task (mean). Decoding uses `temperature=0.6, top_p=1.0, max_new_tokens=32K`, with a 256K context window.*

- *MiniAppBench: A 500-task coding benchmark evaluating whether models can turn a single user request into complete, usable interactive HTML apps in real-world application-generation scenarios. Evaluated with `temperature=1.0, top_p=1.0, max_tokens=128K`.*
- *AntSWEBench: AntSWEBench is an internally used software engineering benchmark that covers mainstream programming languages such as Java, JavaScript, and Python, including various development scenarios like new feature, bug fix, and code refactoring.*
- *Tau3-banking-AA: Aligned with the AA leaderboard, utilizing GPT-5.4-mini (medium reasoning) for both the user simulator and the natural-language assertion judge.*
- *MCP-Atlas: Evaluated on the 500-task public set using the official v1 harness with a 20-turn limit and Gemini-2.5-Pro as the claim-coverage judger.*
- *SkillsBench: Evaluated via kilo-code on 87 tasks (excluding external API-dependent tasks), averaged over 3 runs.*
- *GDPval v2-AA: Evaluated on the public 220-task benchmark using the official Stirrup harness, with a 250-turn limit and a 5-hour timeout.*
- *Search-agent: For all search-agent tasks, evaluations are performed using an internal harness. The basic ReAct paradigm is adopted for single-agent evaluation, while a multi-agent setup is employed for BrowseComp. The reported metric is the average pass@1.*
    - *WideSearch: Evaluated using the official prompt and the official judge model GPT-4.1 on the corrected version of the dataset.*
    - *Draco: Scored based on official rubrics per question, with the final score calculated as the average across all questions using Claude Opus 4.6 as the scoring model.*

- *BrowseComp (Single-Agent): Evaluated using a resume strategy for context management: once the context reaches a 64K-token threshold, the trajectory is summarized, the original history is discarded, and execution is resumed from the summary.*
- *BrowseComp (Multi-Agent): Evaluated using an internal multi-agent search harness based on SearchSwarm/Tongyi DeepResearch, configured with `temperature=0.85, top_p=0.95, max_tokens=8K`, and main/sub-agent context windows of 128K and 64K, respectively.*

Quantized Models

We evaluate the quantized models using several datasets. The FP8 quantized model is applied via the blockwise quantization, and INT4 and FP4 models are applied via groupwise quantization with routed experts weights.

dataset	BF16	FP8	INT4	FP4
GPQA-diamond	84.97	84.00	83.65	82.42
IFBench	74.49	73.40	72.20	72.33
SciCode	41.24	40.37	39.35	39.79
ArcPrize	68.75	67.18	67.56	64.16

Quickstart

Install our SGLang

```
pip install uv

uv venv ~/my_ling_env

source ~/my_ling_env/bin/activate

git clone -b ling_v3_support_mxfp4 https://g

cd sglang_ling_v3

pip install --upgrade pip

pip install -e "python"

pip install flashinfer-cubin==0.6.16.post1 -

pip install flashinfer-python==0.6.16.post1
```

Run Inference

Here is an example to run Ling-3.0-flash with 2 Blackwell GPUs, where the master node IP is `${MASTER_IP}` and server port is `${PORT}`:

Server

Since the model is trained with MTP, we recommend enabling MTP during inference (i.e., `--speculative-algorithm NEXTN`) for lower latency.

```
export SGLANG_ALLOW_OVERWRITE_LONGER_CONTEXT
export SGLANG_JIT_DEEPGEMM_PRECOMPILE=1
export SGLANG_ENABLE_SPEC_V2=1
export FLASHINFER_DISABLE_VERSION_CHECK=1
python -m sglang.launch_server \
    --model-path "$MODEL_PATH" \
    --trust-remote-code \
    --nnodes 1 \
    --dist-init-addr "$MASTER_IP:2345" \
```

```
    --port "$PORT" \
    --tp-size 2 \
    --ep-size 1 \
    --max-running-requests 64 \
    --max-mamba-cache-size 320 \
    --chunked-prefill-size 8192 \
    --allow-auto-truncate \
    --context-length 262144 \
    --random-seed 308534008 \
    --attention-backend trtllm_mla \
    --disable-flashinfer-autotune \
    --mem-fraction-static 0.85 \
    --fp8-gemm-backen cutlass \
    --tool-call-parser ling3 \
    --reasoning-parser ling3 \
    --moe-runner-backend flashinfer_mxfp4 \
    --flashinfer-mxfp4-moe-precision default
    --enable-fp32-lm-head \
    --disable-shared-experts-fusion \
    --speculative-algorithm NEXTN
```

Client

We recommend using the sampling parameters `temperature=0.6`, `top_p=0.95`, and `top_k=20`, and enabling `enable_thinking` for better performance.

```
curl -s http://${MASTER_IP}:${PORT}/v1/chat/
  -H "Content-Type: application/json" \
  -d '{"model": "auto",
       "messages": [{"role": "user", "conten
       "chat_template_kwargs": {"enable_thin
       "stream": true,
       "temperature": 0.6,
       "top_k": 20,
       "top_p": 0.95
     }'
```